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16 July 2025

Re: NCRSH Ethics Application

To Whom It May Concern:

We are pleased to submit the research proposal entitled "Data Collection on Energy Systems and Medical Devices in Healthcare Facilities in Malawi" for ethical review.

The field work will begin in August 2025 and will address the current state of energy supplies, medical devices and equipment in healthcare facilities throughout Malawi.

All documents required for submission, as presented in the checklist are included in this application.

Thank you for considering this application.

Sincerely,

Till Häussner & Gian Teufen

Data Collection on Energy Systems and Medical Devices in Healthcare Facilities in Malawi

Principal Investigators (PI)

Name	Title	Group/Chair/Institute	University
Gian Teufen	Mr.	Global Health Engineering	ETH Zurich
Till Häussner	Mr.	Global Health Engineering	ETH Zurich

Involved Researchers

Name	Title	Group/Chair/Institute	University
Frank Mnthambala	Dr.	Agrisciences Department	Mzuzu University
Hope Kelvin Chilunga	Mr.	Department of Computer Science and Information Technology	Malawi University of Business and Applied Science
Rebecca Alcock	Dr.	Global Health Engineering, Center for Global Engineering	ETH Zurich, University of Toronto
Elizabeth Tilley	Prof. Dr.	Global Health Engineering	ETH Zurich
Jakub Tkaczuk	Dr.	Global Health Engineering	ETH Zurich

Summary

Study Objective: This study aims to address the critical issue of energy access in Malawi's healthcare facilities (HCF). The objective is to collect detailed facility-level data on power supply status, including energy source mix, energy storage, and outage preparation. It will also document the origins and current operations and maintenance (O&M) structures of energy systems and key medical devices. The goal is to create a dataset that enables accountability among stakeholders and provides insights into the successes and failures of current energy systems. Further, data on the medical devices, amenities and equipment present will be collected for better energy load estimations and an updated estimation of the service readiness indicators and index as part of the SARA evaluation tool by WHO. These data will aim to inform decision-making and help prioritize interventions for improving energy access and quality of care in Malawi's HCF.

Methods: Data collection involves field visits of 2-3 HCF per week. Data will be collected through semi-structured questionnaires and observations.

Participants: HCF administrators, management and responsible persons.

Potential risks: The interview might pose a time burden for participants.

Conclusion: This research will generate a detailed dataset on energy access, O&M, medical devices and basic service readiness in Malawian healthcare facilities, addressing a critical information gap. The findings will offer insights into power system functionality and maintenance, as well as more accurate energy load estimations, informing strategic decisions to improve energy access in HCF. This study also aims to enhance stakeholder accountability and strengthen healthcare delivery across Malawi.

1.Background and literature review

Even though the importance of energizing HCF seems natural, data on the topic remains scarce and fragmented (SETI, 2018). This consequently limits the ability of governments, development partners, and service planners to make informed decisions, track progress, or prioritize interventions effectively. The majority of studies to address energy access have focused primarily on household settings, while comprehensive data on healthcare facilities remain limited (Jeuland et al., 2021; SETI, 2018). Work on energizing HCF in Africa is especially rare. Indeed, several African nations do not appear to have a single quantitative peer-reviewed study that investigates an energy-related question (Jeuland et al., 2021). In sub-Saharan Africa, one in four HCF operates entirely without electricity, most of them are located in rural areas. Of the ones that have electricity, three-quarters do not have reliable power supply (Adair-Rohani et al., 2013; Moner-Girona et al., 2021).

Of HCFs in Malawi, 72% have grid access, 21% had other electricity sources such as solar and 7.1% have no access to any electricity at all. The latter are mostly primary-level facilities, particularly rural health posts (World Bank et al., 2023). But access still comes with frequent blackouts disrupting service delivery (Reuland et al., 2020). While renewable energy systems like solar photovoltaic (PV) have been implemented, many are insufficient in capacity and suffer from substandard installation or lack proper maintenance, failing to meet the full energy demands of facilities and leading to frequent load disconnects (UNICEF, 2018). Poor maintenance leads to frequent breakdowns and repairs are often delayed, primarily due to insufficient maintenance staff, supplies, backup energy sources and funding (Franco et al., 2017, p. 20; Shadvar & Rahman, 2024). Load management in Malawian healthcare facilities has not been well addressed, with installations frequently failing to sufficiently consider electrical loads, their distribution, or appliance efficiencies, which results in frequent load disconnects, energy not being matched to priority needs and lower functionality of medical equipment (Ngongoma et al., 2025; Suhlrie et al., 2018).

1.1 Problem Statement

Few studies exist that gather detailed data on power systems, electrical loads, devices and appliances in HCFs in rural Malawi. This hinders governments, development partners, energy engineers and service planners from making data-informed decisions, tracking progress and prioritizing interventions.

A large percentage of power systems and medical devices come from donations. Sometimes the approach to donations seems to be that the upfront financing and short-term O&M are covered, while the facilities are expected to cover long-term costs and responsibility without sufficient institutional or governmental support.

These problems are most prevalent in rural areas, in facilities that provide primary care level services. This includes health posts, dispensaries, maternity clinics, health centers and community hospitals. Primary level facilities account for over 90% of Malawi's health system (World Bank Group, 2019).

1.2 Justification

The study contributes to the growing body of literature on energy access planning in healthcare by analyzing the availability, functionality, regional variation, and management of power systems, alongside the general service readiness, in Malawian healthcare facilities (HCF).

This also provides an opportunity to pilot the proposed SARA energy module (WHO and World Bank, 2015), allowing us to refine and validate this standardized survey tool for measuring energy access in healthcare facilities. Through the pilot phase, we will identify whether we can successfully combine the gathered information into meaningful indicators like the multi-tier electricity access framework and the Index of Electricity Access in Health Clinics (WHO and World Bank, 2015)

The dataset and the resulting policy white paper that provides an overview of successes and failures, offer valuable insights into landscape of ownership, deployment, and maintenance responsibilities and offers a strategy for improving accountability among stakeholders would be valuable for the sector.

2. Project

2.1 Study Objective

Main Objectives

This study aims to collect detailed, facility-level data on power supply status but also on the origins and current O&M structure of energy systems and further take inventory of medical devices and electrical appliances and amenities present. This data will then be used to draw up a policy white paper that provides an overview of successes and failures and offers a strategy for improving accountability among stakeholders moving forward.

Specific Research Questions

1. What proportion of healthcare facilities in Malawi have functioning energy systems, and what types of primary and backup energy sources, energy storage, and outage protection are currently in place?
2. Among energy systems and medical devices, what percentage remain functional, and which stakeholders are accountable for their deployment, ownership, and ongoing maintenance?
3. How do the characteristics of power systems (e.g., energy source mix, functionality, maintenance) differ by region, catchment population size, and associated donors or implementing partners?
4. Does the HCF have an established plan to reduce or reorganize their loads if not all demand can be met?
5. What is the estimated service readiness index as defined by SARA of WHO for the facilities?
6. How does the energy situation of facilities correlate with their overall service readiness index?
7. Is there already a noticeable influence of climate change on the operations in HCFs?
8. What medical devices and electrical appliances are present in HCFs and what are their characteristics, functionality and actual power consumption?
9. How can energy loads be estimated based on the electrical devices present and their power consumption?

Hypothesis

1. A majority of primary care facilities in Malawi lack consistent access to functioning power systems, relying primarily on the national grid or solar energy installations as the main sources and diesel or petrol generators as backup, with limited energy storage capacity and minimal outage protection mechanisms.
2. A substantial portion of installed energy systems and essential medical devices in Malawi's primary care facilities are either damaged or broken, due to O&M difficulties.

3. Using detailed device information and measured power consumption values, electrical loads of the facilities can be estimated more accurately than using standard values from literature.
4. Power system characteristics vary significantly across regions and stakeholders.
5. Most HCFs do not have a formalized load management plan in place, resulting in the inability to prioritize critical loads during power shortages.
6. HCFs with greater access to reliable and adequate energy will exhibit higher overall service readiness.
7. Climate change is already having a noticeable impact on the operations of healthcare facilities in Malawi, primarily through increased frequency of power outages, heat-related strain on staff and patients, though we suspect staff do not necessarily make a direct connection between these realities and climate change

2.2 Methods and Study Design

Method of data collection

Inspection

Qualitative and Quantitative data will be collected on various characteristics of power supply status, including the energy source/storage mix, nameplate characteristics and outage preparation and protection. Accessing this information will require talking to a responsible person; however, the collecting of nameplate characteristics and other related energy metrics can be conducted by the PI without additional support once pointed in the right direction. Categories are based and combined on the ones provided by various sources to ensure full coverage and international compatibility (WHO, 2015a; Ngongoma et al., 2025; Reuland et al., 2020; Suhlrie et al., 2018).

Photographic documentation of energy system components and medical equipment on facility premises will be conducted. These images will solely capture equipment status and will strictly exclude any identifiable individuals or patient information.

Structured Questionnaire

The questionnaire is composed of these sections:

0) Facility Identifiers
1) Energy System
2) Origins, Operations and Maintenance
3) Climate Change & Weather Impact
4) Amenities & Equipment

In section zero the facility identifiers are collected. This includes information on location, facility type, managing authority and patient count.

Section one is the entirety of the “SARA energy module: sample survey questions for multi-tier measurement” proposed in 2015 (WHO and World Bank, 2015). It was specifically designed to ensure that all the questions needed for international analyses are covered. It includes all of the energy related questions proposed in the harmonized health facility assessment (HHFA) (WHO, 2023) but offers additional more in-depth questions about electric capacity, evening supply, financing and more. This

study also serves as a pilot for the module and an according review of the experience with it will be included in the report.

In the second section, extensive qualitative data is collected on current O&M practices, the original project structure and financing mechanisms of the power supply.

The third section explores the connection between climate change and HCF. Whether there has already been any occurrence of extreme weather conditions and consequently other noticeable changes in operations or patient population.

The fourth section covers the questions needed to determine the general service readiness indicator as proposed by SARA (WHO, 2015). This involves assessing five key domains: basic amenities, basic equipment, standard precautions for infection prevention, diagnostic capacity, and essential medicines. Each of those indicators is given a score based on reported availability, out of which the main index score is calculated, ensuring a holistic evaluation of a health facility's fundamental operational capabilities.

The full questionnaire is provided in Appendix 3.

Study Design

Study Site

The planned location timeline for the thesis is in the order north, south and central, with a one-month stay each in Mzuzu, Blantyre and Lilongwe. Per week, about 2-3 health posts will be visited and the device inventory will be logged. Planning for the individual facility visits will be done later in Malawi to account for schedule changes and anything unforeseen.

Study Visits

Study visits will be announced in collaboration with the respective DHOs in advance. On arrival, all staff and patients present will be informed about the study using a quick oral presentation and the written handouts in both English and Chichewa.

Participants will then be recruited for the questionnaire and will sign the information and consent form.

The medical device and electrical appliance inventory will be studied by the researchers themselves with help of staff without a dedicated questionnaire. Details on devices will include name, brand, type, nameplate energy characteristics and functionality. Pictures of devices will be taken and device power consumption will be measured using standard household energy loggers. This will be done as nonobstructive as possible to minimize negative impact on facility workings and healthcare provisions. Management staff and participants will at any point be able to pause or stop the research in the facility without giving any reason or receiving any disadvantages to always ensure proper facility workings.

2.3 Participants for Questionnaire

Number of Participants

Per facility, 1-2 knowledgeable persons will be questioned on-site. With an estimated 40 facilities to be visited this would amount to about 60 participants.

Inclusion Criteria

- Individuals who are knowledgeable about the energy system of the HCF
- Individuals aged more than 18 years
- When available, individuals who have worked on the site for more than two years

Exclusion Criteria

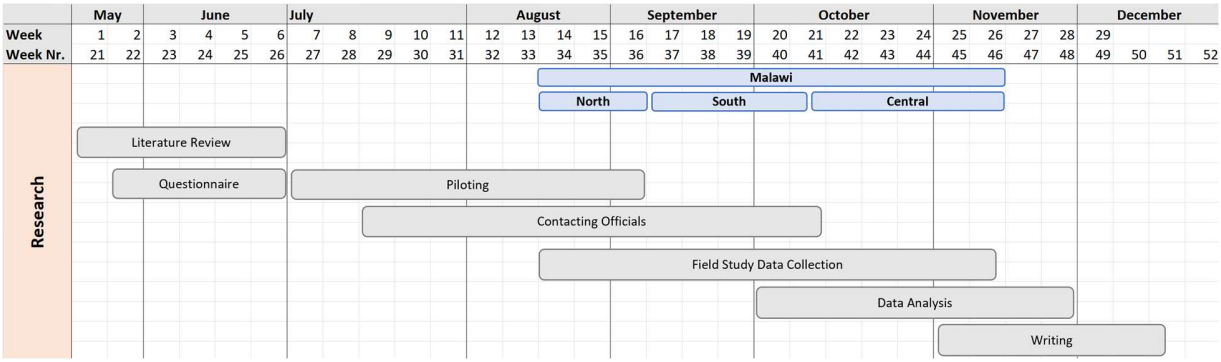
- Individuals who do not consent to being interviewed

Recruitment Procedure

In consultation with the respective DHOs, management and administrative contacts of the respective healthcare facilities will be identified and informed about the study. With facility management, fitting survey participants will be recruited for participation based on criteria mentioned above.

2.4 Project Timeline

The research will commence in August in facilities on northern Malawi and Mzuzu. The researchers will then move to Blantyre for September and conduct research in facilities in southern Malawi and finally conduct research in districts in central Malawi based out of Lilongwe in October.



2.5 Project Partners and Funding

Source of Funding

The project is funded by the Global Health Engineering (GHE) group at ETH Zurich.

Roles and Rights of Project Collaborators

- Rebecca Alcock (Dr., Global Health Engineering, ETH Zurich; Center for Global Engineering, University of Toronto): She is the main supervisor of the thesis and will have access to the project data for providing input if needed.
- Elizabeth Tilley (Prof. Dr., Global Health Engineering, ETH Zurich): She provides overall supervision in the GHE group, with access to the project data.
- Jakub Tkaczuk (Dr., Global Health Engineering, ETH Zurich): He is the second supervisor of the thesis and will have access to the project data for providing input if needed.
- Frank Mnthambala (Dr., Agrisciences Department, Mzuzu University): He is our contact for advice in Mzuzu.

- Hope Kelvin Chilunga (Mr., Lecturer at Malawi University of Business and Applied Science): He is our contact for advice in Blantyre.

The research group GHE brings an extensive network of collaborators and experience in Malawi, having conducted numerous studies on topics of health, renewable energy, solid waste management and more within the country. This understanding of the local context and existing research helps our current study to be as nuanced and effective as possible.

Detailed Budget and Justification

Item	Unit	Cost/unit (USD)	Nr. of units	Total cost (USD)	comments
Flipcharts	**Number of flipcharts**	**2**	**2**	**4**	
Plastic Documents Sleeves	**Number of plastic document sleeves**	**1**	**10**	**10**	
Printed Info Handouts	**Number of printed info handouts**	**0.1**	**100**	**5**	**50 English copies and 50 Chichewa copies**
Printed Consent Forms	**Number of printed info & consent forms**	**0.1**	**160**	**8**	**80 English copies and 80 Chichewa copies**
Energy meter	**Number of Energy Meters**	**60**	**2**	**120**	
SD cards	**Number of SD Cards**	**20**	**3**	**60**	
Car	**Number of months renting the car**	**500**	**4**	**2000**	
Gas	**Number of months**	**500**	**4**	**2000**	

PPE Masks	**Number of masks**	**1**	**20**	**20**	
Disinfection bottles for health posts	**Number of disinfection bottles**	**2**	**50**	**100**	
			Total cost (USD	**4327**	
Ethics Application			**10% of total cost (USD)**	**433**	
			Total project cost (USD)	**4760**	

3. Ethical Aspects

3.1 Informed Consent and Debriefing

Recruitment of Participants

With the approval of the Malawian "National Committee On Research In The Social Sciences" (NCRSH), the district health offices (DHO) will be individually contacted, informed about the contents of our study and asked for contacts to facility administrators in their respective districts. With them we will communicate in advance and schedule an appropriate date for us to visit. They will be asked to sign the consent form before the interview.

Dissemination of Study Information

Detailed study information will be sent in advance to each DHO and facility in the form of this ethics application. Each participant will also be explained the structure of the interview individually by briefing the consent form in Appendix 1.

Expression of Informed Consent

Participants will express their informed consent by signing the consent sheet (Appendix 1). The research team will work with the facility administrator to ensure compliance with ethical guidelines and the participant gets to keep a copy of the consent form.

Storage of Consent Forms

The consent forms will be retained by the GHE group for 5 years for record-keeping and verification. These will be stored in a locked cabinet. Only Prof. Dr. Tilley will have access to the keys.

Consent for Illiterate Participants

For participants who are illiterate, informed consent will be recorded using a fingerprint. A witness will be present to verify and record the consent.

Sharing Study Results

As soon as everything is digitized, the collected data of each individual facility will be sent to the facility administrator and relevant facility data to the corresponding DHO. Once the thesis is completed and published to the ETH research collection the report will be sent to each DHO and facility.

3.2 Data Protection and Publication

Data Collection

Individuals participate anonymously. No personal data will be collected. The data will only be affiliated with the facility. Study questionnaire data will be collected digitally into tablets/smartphones using (KOBO Toolbox) software.

Data Storage

Consent forms and any other paper documents will be stored in a locked cabinet in the study office. Only the project lead and Prof. Tilley from the Global Health Engineering research group will have access to these documents.
Study tablets/smartphones will be backed up to a study PC and data deleted from the handheld devices every alternate day. Data will be backed up to the ETH internal server for storage.

Data Analysis

R Statistical Software and RStudio IDE will be used for quantitative data analysis to generate the results in the manuscript. The raw as well as the aggregated processed data will be published as an R package. Individual participants cannot be identified. The analysis code that will be used to generate the figures and tables of the manuscript will be contained in a public GitHub repository.

Anonymization

All data will be completely anonymous to the facility level, instead of individual level.

Publication of Results

- **Forms:** Results will be published in research papers, blogs, and policy briefs to disseminate findings widely and inform both scientific and public communities.
- **Accessibility:** Data will be made accessible in an anonymous form, such as through a repository, to allow for further research and verification.

3.3 Compensation

Participants will not receive monetary compensation for their participation in the study.

3.4 Risk and Discomforts

Participating in the study entails no physical or psychological risks or discomforts. The interview might pose a time burden for participants. If participants wish to interrupt the interview, they can do so at any time. There are also no social or environmental risks in the future.

3.5 Ethics Reviews by funding institution ETH Zurich

The questionnaire was approved by the Ethics Commission of ETH with number *25 ETHICS-206*. The data collection on the medical device and appliance inventory was given a separate waiver with number *EK-2025-E-19*. Both documents are available on request.

4. References

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Appendix 1a: Facility Info Sheet English

ETH zürich | Mzuzu University – Service Perfection Self-Reliance | MALAWI UNIVERSITY of BUSINESS AND APPLIED SCIENCES

Research survey:

Energy and Device Data Collection in Healthcare Facilities in Malawi

A survey on energy supply systems, maintenance & operations and the medical device & equipment inventory

Principal Investigators & Contact:

Name	Group/Chair/Institute	University	Email	Phone
Mr. Gian Teufen	Global Health Engineering	ETH Zurich	teufeng@ethz.ch	+41 76 421 31 22
Mr. Till Häussner	Global Health Engineering	ETH Zurich	thaeussner@ethz.ch	+41 78 755 10 40

Involved Researchers:

Name	Group/Chair/Institute	University
Dr. Frank Mnthambala	Agrisciences Department	Mzuzu University
Mr. Hope Kelvin Chilunga	Department of Computer Science and Information Technology	Malawi University of Business and Applied Science
Dr. Rebecca Alcock	Global Health Engineering, Center for Global Engineering	ETH Zurich, University of Toronto
Prof. Dr. Elizabeth Tilley	Global Health Engineering	ETH Zurich
Dr. Jakub Tkaczuk	Global Health Engineering	ETH Zurich

What is investigated and how?

This study addresses energy systems and the medical device and equipment inventory in healthcare facilities in Malawi. Data collection involves visits to approximately forty facilities throughout the country. The goal of this study is to create a public and freely accessible dataset of detailed energy system status and the medical device inventory functionality. This information will help us understand what medical devices are functional, how much energy the facility needs, what organization is responsible for the energy systems, and what works well and what doesn't.

What data is collected?

- Data on the facility like amenities, devices, equipment, infrastructure, energy system and maintenance is collected and photos thereof will be taken.
- No personal data of any kind will be collected.

What will the data be used for and how will it be published?

The data will be used for accurate modeling of energy needs of healthcare facilities, to understand the operations and maintenance of energy systems and in the future for scaling standalone solar microgrids for rapid facility electrification. Further, we'll use data on facility amenities and equipment for an estimation of the service readiness indicators and index as part of the SARA evaluation tool by WHO. Data will be published openly in a free database and shared with all participating facilities and corresponding DHOs.

Who is funding the study?

Prof. Dr. Elizabeth Tilley of Global Health Engineering at ETH Zurich, Switzerland.

Who reviewed and approved the study?

- ETH Zurich Ethics Commission in Switzerland
- National Committee On Research In The Social Sciences and Humanities (NCRSH) in Malawi

More information:

Global Health Engineering at ETH Zurich: **ghe.ethz.ch**

Thank you for your cooperation!

Appendix 1b: Facility Info Sheet Chichewa

Kafukufuku: Kusonkhanitsa Zambiri pa Mphamvu ndi Zipangizo mu Malo Azachipatala ku Malawi

Kafukufuku pa machitidwe a magetsi, kukonza ndi ntchito zake komanso mndandanda wa zipangizo ndi zida zamankhwala.

Ofufuza Oyamba ndi Zolumikizirana:

Dzina	Gulu / Dipamenti	Yunivesite	Imelo	Foni
Mr. Gian Teufen	Global Health Engineering	ETH Zurich	teufeng@ethz.ch	+41 76 421 31 22
Mr. Till Häussner	Global Health Engineering	ETH Zurich	thaeussner@ethz.ch	+41 78 755 10 40

Ofufuza Omwe Akutenga Nawo Mbali:

Dzina	Gulu / Dipamenti	Yunivesite
Dr. Frank Mnthambala	Agrisciences Department	Mzuzu University
Mr. Hope Kelvin Chilunga	Department of Computer Science and Information Technology	Malawi University of Business and Applied Science
Dr. Rebecca Alcock	Global Health Engineering, Center for Global Engineering	ETH Zurich, University of Toronto
Prof. Dr. Elizabeth Tilley	Global Health Engineering	ETH Zurich
Dr. Jakub Tkaczuk	Global Health Engineering	ETH Zurich

Kafukufuku Akufuna Kudziwa Chiyani ndi Motani?

Kafukufukuyu akufuna kudziwa za machitidwe a mphamvu zamagetsi ndi mndandanda wa zipangizo zamankhwala m'malo azachipatala ku Malawi. Zambiri zidzasonkhanitsidwa mwa kukaona zipatala pafupifupi 40 m'dziko lonselo. Cholinga chake ndi kupanga fayilo yaulere yomwe ili ndi zambiri za momwe mphamvu zamagetsi zilili komanso momwe zipangizo zamankhwala zikuyendera. Izi zithandiza kudziwa zipangizo zomwe zikugwira ntchito, kuchuluka kwa mphamvu zomwe chipatalacho chimalandira, bungwe loyang'anira mphamvu zamagetsi, komanso zinthu zomwe zikuyenda bwino ndi zomwe sizikuyenda.

Deta Zotengedwa Zidzakhala Zotani?

- Zambiri zokhudza chipatala monga zinthu zomwe zilipo, zipangizo, zomangamanga, makina amagetsi, ndi kukonza zidzasonkhanitsidwa. Zithunzi za zinthu izi zidzatengedwa.
- Palibe zambiri za munthu payekha zomwe zidzasonkhanitsidwe.

Deta Yidzagwiritsidwa Ntchito Bwanji Ndipo Zidzatulutsidwa Motani?

Zambiri zidzagwiritsidwa ntchito pakupanga chitsanzo chenicheni cha zosowa zamagetsi za zipatala, kudziwa mmene makina amagetsi amagwirira ntchito, komanso pakukonza mapulani okulitsa magetsi a dzuwa (solar microgrids) m'tsogolo. Zidzagwiritsidwanso ntchito pa kuwunika zokonzekera kupereka chithandizo pogwiritsa ntchito chida choweluza cha SARA cha WHO. Zambiri zidzagulitsidwa pa intaneti mwaulere ndipo ziperekedwa kwa zipatala zonse zomwe zatenga nawo mbali komanso ma DHO awo.

Ndani Akuthandiza Kafukufukuyu?

Prof. Dr. Elizabeth Tilley wa Global Health Engineering ku ETH Zurich, Switzerland.

Ndani adawunika ndi kuvomereza kafukufukuyu?

- Komiti ya ETH Zurich Ethics ku Switzerland
- National Committee on Research in the Social Sciences and Humanities (NCRSH) ku Malawi

Zambiri Zowonjezera muzipeza ku:

Global Health Engineering ku ETH Zurich: **ghe.ethz.ch**

Zikomo Potithandiza!

Appendix 2a: Consent Form English

ETH zürich Mzuzu University Service Perfection Self-Reliance MALAWI UNIVERSITY of BUSINESS AND APPLIED SCIENCES

Information and consent form

Energy and Amenity Data Collection in Healthcare Facilities in Malawi

Place and time:

Contact person if you have any questions:	Gian Teufen, Global Health Engineering (GHE) ETH Zurich, teufeng@ethz.ch, +41 76 421 31 22
Data Protection Officer ETH Zurich:	Tomislav Mitar (tomislav.mitar@sl.ethz.ch);
NCRSH Malawi:	ncrsh@ncst.mw

We would like to ask you if you are willing to participate in our research project. Your participation is voluntary. Please read the text below carefully and ask the conducting person about anything you do not understand or would like to know.

What is investigated and how?

This study addresses energy systems and the medical device and equipment inventory in healthcare facilities in Malawi. Data collection involves visits to approximately forty facilities throughout the country. The goal of this study is to create a public and freely accessible dataset of detailed energy system status and the medical device inventory functionality. This information will help us understand what medical devices are functional, how much energy the facility needs, what organization is responsible for the energy systems, and what works well and what doesn't.

Who can participate?

People at least 18 years old, who know about the energy system at their healthcare facility and have ideally worked there for more than two years.

What am I supposed to do as a participant?

You will be guided through a questionnaire about the facility. Then, we will ask you to show us the energy systems and some medical equipment. The investigators may also take photographs of energy systems or medical devices on the premises to document their condition.

What are my rights during participation?

Your participation in this study is voluntary. You can withdraw your participation at any time without giving reasons and without any disadvantage.

What risks and benefits can I expect?

The interview might take some of your time. However, your help will allow the government, aid groups, and planners to make better decisions and improve energy and resources for healthcare facilities in the future.

Will I be compensated for my participation?

No, we won't provide financial compensation.

What data is collected from me and how is it used?

The following data is collected from you:

- No personal data is collected from anyone.
- Data on the facility like amenities, devices, equipment, infrastructure, energy system and maintenance is collected and photos thereof will be taken.

Purpose of Data:

The data will be used for accurate modeling of energy needs of healthcare facilities, to understand the operations and maintenance of energy systems and in the future for scaling standalone solar microgrids for rapid facility electrification. Further, we'll use data on facility amenities and equipment for an estimation of the service readiness indicators and index as part of the SARA evaluation tool by WHO. This will provide an up-to-date nationwide data set on the status of small healthcare facilities.

Handling of Identifying Data:

We will not collect any personal information that can identify you, so you will remain completely anonymous. This paper consent form with your name on it will be stored in a locked cabinet at ETH Zurich for five years, accessible only by Prof. Dr. Tilley.

Publication

Our findings will be shared in research papers and easier-to-read formats like blogs and reports. No one will be able to identify you from anything we publish. The anonymous dataset will be published openly for further use as an R package and a public GitHub repository for analysis code allowing other researchers to build upon our findings.

The collected data of your facility and the full national analysis will be shared with the facility administrator and the corresponding District Health Office (DHO) once digitized.

Who is funding the study?

The study is funded by the Global Health Engineering (GHE) group at ETH Zurich, Switzerland

Who reviewed the study?

This study was reviewed and approved by the Malawi Ethics Commission (XX) and the ETH Zurich Ethics Committee under application number 25 ETHICS-206.

Complaints office

The secretariat of the ETH Zurich Ethics Committee is available to help you with complaints in connection with your participation in the study. Contact: ethics@sl.ethz.ch or 0041 44 632 85 72.

Consent Form

I, the participant, confirm with my signature that:

- I have read and understood the study information.
- I had enough time to decide about my participation and am taking part in the study voluntarily.
- I fulfil the stated conditions for participation.
- I consent that the data described above may be collected from me and used as described.
- I know that the data collected from me may be published in anonymized form.
- I know that I can cancel my participation any time.

I would like to be informed about the results of this study

☐ Yes, name, phone number or e-mail: .....

☐ No

Name of participant

.....

Place, date

.....

Signature participant

.....

Place, date

.....

Signature conducting person

.....

Appendix 2b: Consent Form Chichewa

ETH zürich Mzuzu University Service Perfection Self-Reliance MALAWI UNIVERSITY of BUSINESS AND APPLIED SCIENCES

Fomu ya Chidziwitso ndi Kuvomereza

Kusonkhanitsidwa kwa deta ya mphamvu ndi zothandizira ku malo othandizira zaumoyo ku Malawi

Malo ndi Tsiku:

Munthu Wothandizira Ngati Muli Ndi Mafunso: Gian Teufen, Global Health Engineering (GHE)
ETH Zurich, teufeng@ethz.ch, +41 76 421 31 22

Woyang'anira Zachinsinsi ku ETH Zurich: Tomislav Mitar (tomislav.mitar@sl.ethz.ch);

Komiti ya Malawi Ethics, NCRSH Malawi: ncrsh@ncst.mw

Tikufuna kukufunsani ngati mungakhale okonzeka kutenga nawo mbali pa kafukufuku wathu. Kutenga nawo mbali ndi kufuna kwanu. Chonde werengani bwino nkhaniyi ndipo funsani munthu wochititsa kafukufukuyu ngati pali china chomwe simukumvetsa.

Kafukufukuyu akufuna kudziwa chiyani ndipo afufusa bwanji?

Kafukufukuyu akufuna kudziwa za machitidwe a mphamvu zamagetsi ndi zipangizo zamankhwala makina a m'zipatala zing'onozing'ono ku Malawi. Zambiri zidzasonkhanitsidwa mwa kukaona zipatala pafupifupi 40 m'dziko lonselo. Cholinga chake ndi kupanga fayilo yaulere yomwe ili ndi zambiri za momwe mphamvu zamagetsi zilili komanso momwe zipangizo zamankhwala zikuyendera. Izi zithandiza kudziwa zipangizo zomwe zikugwira ntchito, kuchuluka kwa mphamvu zomwe zipatala zikufuna, bungwe loyang'anira mphamvu zamagetsi, komanso zinthu zomwe zikuyenda bwino ndi zomwe sizikuyenda.

Ndani angatenge nawo mbali?

Anthu azaka 18 kupitirira amene amadziwa bwino za mphamvu zamagetsi ku chipatala chawo ndipo, ngati n'kotheka, agwirepo ntchito kwa zaka zopitirira ziwiri.

Kodi ndichite chiyani ngati ndatenga nawo mbali?

Mudzadzazitsidwa funsofuno lokhudza chipatala. Kenako mudzafunsidwa kuti muwonetse makina amagetsi ndi zipangizo zamankhwala. Ofufuza akhoza kutenganso zithunzi za makina amagetsi kapena zipangizo zomwe zili mu chipatalacho.

Ufulu Wanga Pa Kafukufukuyu ndi otani?

Kutenga nawo mbali ndi kwa kufuna kwanu. Mutha kusiya nthawi iliyonse osapereka chifukwa ndipo popanda chilango chilichonse.

Ndiyembekezere zowopsa ndi zopindulitsa zotani?

Mafunso akhoza kutenga nthawi yanu. Komabe, thandizo lanu lithandiza boma, mabungwe othandiza, ndi oyang'anira mapulojekiti kupanga zisankho zabwino ndiponso kusintha mphamvu ndi zinthu zofunikira zipatala zikufuna mtsogolo.

Kodi ndidzalandira malipiro chifukwa chotenga nawo mbali?

Ayi, palibe ndalama zomwe zidzaperekedwe chifukwa chotenga nawo mbali.

Kodi deta yotengedwa kuchokera kwa ine ndi yotani ndipo imagwiritsidwa ntchito bwanji?

Deta Yotengedwa kea inu:

- Palibe zambiri zanu zanu zomwe zidzasonkhanitsidwe.
- Zomwe zidzasonkhanitsidwe ndi zambiri zokhudza chipatalacho monga zinthu zothandizira, zipangizo, zomangamanga, makina amagetsi, ndi kasamalidwe. Zithunzi za zinthu izi zidzatengedwa.

Cholinga cha Deta:

Zambiri zidzagwiritsidwa ntchito pa kupanga chithunzi chenicheni cha zosowa zamagetsi za zipatala, kudziwa mmene makina amagetsi amagwirira ntchito, komanso pakukonza mapulani okulitsa magetsi a dzuwa (solar microgrids) m'tsogolo. Zidzagwiritsidwanso ntchito pa kuwunika zokonzekera kupereka chithandizo pogwiritsa ntchito chida choweluza cha SARA cha WHO. Zambiri zidzagulitsidwa pa intaneti mwaulere ndipo ziperekedwa kwa zipatala zonse zomwe zatenga nawo mbali.

Kusamalira kwa Deta:

Sitingatengereko chilichonse chomwe chingakuzindikiritseni. Fomu yovomerezayi yokha ndi yomwe ili ndi dzina lanu ndipo idzasungidwa mu kabati yotsekedwa ku ETH Zurich kwa zaka zisanu, ndipo Prof. Dr. Tilley yekha ndiye adzakhala ndi mwayi kuziwona.

Kufalitsa

Zotsatira zathu zidzagawidwa mu mabuku a kafukufuku komanso m'njira zosavuta monga malipoti. Palibe amene adzadziwe kuti ndi inu kuchokera mu zotsatira zomwe zidzatulutsidwe. Deta yosadziwika yomwe imasonyeza chipatala chanu idzagawidwa mwaulere monga R package komanso pa GitHub ya anthu onse. Deta yokhudza chipatala chanu komanso malipoti a dziko lonse adzaperekedwa kwa oyang'anira chipatalacho ndi ku DHO akamalizidwa.

Akuthandiza Kafukufukuyu ndi ndani ?

Kafukufukuyu akuthandizidwa ndi gulu la Global Health Engineering ku ETH Zurich, Switzerland.

Ndani Anawunika Ndi Kuvomereza Kafukufukuyu?

Kafukufukuyu wawunikidwa ndi kuvomerezedwa ndi Malawi Ethics Commission (*review number*) ndi ETH Zurich Ethics Commi ee pansi pa applica on number 25 ETHICS-206.

Ofesi Yovomereza Madandaulo:

Secretariat ya ETH Zurich Ethics Committee ilipo kuthandiza ndi madandaulo okhudzana ndi kafukufukuyu. Lumikizanani: ethics@sl.ethz.ch kapena +41 44 632 85 72.

Fomu Yovomereza

Ine, wotenga nawo mbali, ndikutsimikizira ndi siginecha yanga ku:

- Ndawerenga komanso ndamvetsetsa zambiri za kafukufukuyu.
- Ndinalandira nthawi yokwanira yoiganizira, ndipo ndikuchita izi mwa kufuna kwanga.
- Ndikukwaniritsa zofunikira zofunikira ku nditenge nawo mbali.
- Ndikuvomera ku zomwe zatchulidwa zitha kutengedwa kuchokera kwa ine.
- Ndikudziwa ku zomwe zidzasonkhanitsidwe zidzamasulidwa mopanda dzina langa.
- Ndikudziwa ku nditha kuyimitsa kutenga nawo mbali nthawi iliyonse.

Ndikufuna ndikudziwitsidwa zotsatira za kafukufukuyu

☐ Inde, dzina, nambala ya foni kapena imelo: .....

☐ Ayi

Dzina la wotenga nawo mbali

.....

Malo, tsiku

.....

Siginecha ya wotenga nawo mbali

.....

Malo, tsiku

.....

Siginecha ya munthu ochititsa kafukufukuyu

.....

Appendix 3a: Questionnaire English

English

Facility Identifiers		
Question	**Logic**	**Choices**
What is the name of the facility?		[Free text]
What is the address or description of the facility location?		[Free text]
Elevation (m)		__ __ __ __
Latitude		N/S __ DEGREES __ __ DECIMAL __ __ __ __
Longitude		N/S __ DEGREES __ __ DECIMAL __ __ __ __
In which region/province is the facility located?		North Central South
In which district is the facility located?		[List of all districts]
What type of facility is this?		National referral hospital District/Provincial hospital Health centre/clinic Health post Maternal/child health clinic Other (specify)
What is the managing authority of the facility?		Government/Public NGO/Not-for-profit Private-for-profit Mission/faith-based Other (specify)
Is the facility located in an urban or rural area?		Urban Rural
Is the facility outpatient only?		Yes No Don't know
On average, how many patients visit the facility per day?		0 1-10 11-20 21-50 51-100 >100
On average, how many patients visit the facility per week?		0 1-50 51-100 101-200

		201-500 >500
Which days of the week is the facility open to patients?	Check all that apply	Monday Tuesday Wednesday Thursday Friday Saturday Sunday
Which days are the busiest per week on average?	Check all that apply	Monday Tuesday Wednesday Thursday Friday Saturday Sunday
How long do outpatients stay in the clinic?		<1 hour 1-2 hours 2-4 hours 4-6 hours >6 hours Don't know
How long is the average consultation?		<5 minutes 5-15 minutes 15-30 minutes 30-60 minutes >60 minutes Don't know
How do patients get to the clinic?	Check all that apply	Foot Bicycle Car (private/taxi) Bus (public transport) Shuttle (facility-provided) Other (specify)
How does the patient population vary over the year? Is there a busy season/which season is the busiest?	Check all that apply	Hot & Wet Season (November - April) Cool & Dry Season (May - August) Hot & Dry Season (September - October) No significant variation

Energy System; SARA Energy Module (WHO, 2015)		
Question	**Logic**	**Choices**
Questions on capacity		
Does your facility have electricity from any source (e.g. grid, generator, solar, or other)?	Skip	Yes No

What is the electricity used for in the facility?	${electricity} = 'no' Check all that apply	Only stand-alone electric medical devices/appliances (e.g. EPUI cold room, suction apparatus, refrigerator, etc.) Electric lighting (excluding flashlights) and communications Electric lighting, communications, and 1 to 2 medical devices/appliances All electrical needs of the facility All electrical needs of the facility and staff housing
If the facility has no electricity, what is its primary lighting source?	Check all that apply	No source of lighting Kerosene lamps Candles Battery-operated lamps/flashlights Solar lanterns Headlamps Other (Specify)
If the facility has electricity, what is its main source? Note: This means the electricity source that you rely upon most of the time.	${electricity} = 'yes'	Central supply (national/community grid) Local mini-grid (i.e., the facility shares electricity supply, typically a renewable source or a generator, with other nearby buildings) Specify the supply modality if known. On-site hybrid solar/generator On-site generator (with battery storage) On-site generator (no battery storage) On-site solar system (except lanterns) Dedicated hydropower system Dedicated wind power system Other (Specify)
Does the facility have an energy storage system like a rechargeable battery system?		Yes No Don't know
Other than the main or primary source, does the facility have a backup source of electricity? If so, what is the secondary source of electricity?	${electricity} = 'yes' Check all that apply	No back-up source Central supply electricity grid Local mini-grid (i.e., the facility shares electricity supply, typically a renewable source or a generator, with other nearby buildings) Specify the supply modality if known. On-site generator (with battery storage) On-site generator (no battery storage) On-site solar system (except lanterns) Dedicated hydropower system Dedicated wind power system Solar lanterns Other (Specify)
If you have both a primary and a secondary/backup source, do you have an energy management system to switch automatically?	${secondary} = 'yes'	Yes, automatic shift, with fuel efficiency optimization (for example with the wind/solar) Yes, automatic shift; but no efficiency optimization No, a manual startup is needed
If you have no backup electricity source, what is your backup lighting source?	${secondary} = 'no'	No back-up source of lighting Kerosene lamps

		Candles Battery-operated lamps/flashlights Headlamps Solar lanterns Other (Specify)
Generator Functionality		
Does the generator have a charged battery available today?	selected(${generator})	Yes No Don’t know
Does the generator have fuel available today?	selected(${generator})	Yes Petrol Diesel No Don't know
Solar System Functionality		
Is the solar system functional?	selected(${solar})	Yes, functioning as expected Partially Battery needs servicing or replacement Specify all parts No, not functional Don’t know
For how many hours per day, on average, does the solar system provide power?	selected(${solar})	4 hours or less 5 to 8 hours 9 to 16 hours 17 to 23 hours (energy storage system required) 24 hours (energy storage system required)
Duration		
On average, how many hours per day is this facility open?		4 hours or less 5 to 8 hours 9 to 16 hours 17 to 23 hours 24 hours
On average, for how many hours in a 24-hour day is electricity available should you need to use it?		4 hours or less 5 to 8 hours 9 to 16 hours 17 to 23 hours 24 hours
Evening supply		
On average, for how many hours during the evening (after 6 p.m.) and night is the facility open for services?		Never open after 6 p.m Until 8 p.m Until 10 p.m. Until midnight All night long
On average, for how many hours during the evening (after 6 p.m.) and night is electricity available at the health facility?		Less than 1 hour 1 to 2 hours 3 to 4 hours 5 to 8 hours 8 to 14 hours

Reliability		
During the past 7 days, was electricity available at all times when it was needed from the main or any backup source when the facility was open for regular or emergency services?		Always available (no interruptions) Often available (some interruptions of less than 2 hours per day) Sometimes available (frequent or prolonged interruptions of more than 2 hours per day)
If connected to the grid, on an average day, how many times does the facility face electricity supply interruptions from the grid?		No interruptions 1-3 interruptions per day More than 3 interruptions per day
If primarily using on-site generation, on an average day, how many times does the facility face electricity supply interruptions?		No interruptions 1-3 interruptions per day More than 3 interruptions per day
On an average day, what is the total duration of all supply interruptions taken together?		No interruptions Less than 30 min 31-60 min 61 min to 4 hours 5-8 hours More than 8 hours
When are supply interruptions most common?		Daytime (before dark) Evening (after dark to midnight) Overnight (midnight to 8 a.m.)
Quality		
Does the facility usually face a problem of low voltage or voltage fluctuations from the main source?		Yes No Don't know
Have voltage fluctuations damaged equipment in the past year?		No Some equipment damaged Significant equipment damaged
Financial Sustainability		
If you use electricity from the grid, did you have sufficient budgetary allocation (funds available) for paying your electricity bills in the last two months?	selected(${grid})	Yes No Don't know Not applicable
If you use electricity from the grid, who pays the electricity bill for the health clinic?	Check all that apply	Ministry of Health/district health office Donor Health facility Combined funds from donor, facility, ministry Patients Don't know
Over the last two months, did you have sufficient budgetary allocation (funds available) for fuel for the stand-alone generator to operate at all times when it was needed?	selected(${generator})	Yes No Don't know
Who pays for generator fuel?	selected(${generator}) Check all that apply	Ministry of Health/district health office Donor Health facility Combined funds from donor, facility, ministry All electricity is metered and billed to patients

		Some electricity is billed to patients Don't know
Over the last year, did you have sufficient budgetary allocation (funds available) to keep the solar panels functional?	selected(${solar})	Yes No Don't know
If you use a solar energy-based electricity solution, who pays for light bulbs and solar parts replacement?	selected(${solar}) Check all that apply	Ministry of Health/district health office Donor Health facility Patients Combination of facility, ministry, donor sources Private payments by staff or patients Combination of private purchases and other sources No one Don't know
Over the last two months, did you have sufficient budgetary allocation (funds available) for maintenance of electricity supply equipment?	${electricity} = 'yes'	Yes No Don't know
Environmental Sustainability		
Does your primary source of lighting (including kerosene lamps or candles) or electricity cause any pollution of air in the local area in or around the health clinic, through production of fumes or gases that potentially harm the health of the patients or the health workers?		Yes No Don't know
Does your primary source of lighting or electricity cause any pollution of water or soil in the local area in or around the health clinic, through improperly discharged oils, exhausts or other pollutants that potentially harm the health of the patients or the health workers?		Yes No Don't know
Does your primary source of electricity cause any significant noise pollution in the local area in or around the health clinic that potentially harms the health of the patients or the health workers?		Yes No Don't know

Origins, Operations and Maintenance		
Question	**Logic**	**Choices**
Origins	${electricity} = 'yes'	
Who financed the original installation of the primary energy system at the facility?		Ministry of Health/district health office Christian Health Association Donor Health facility Combined funds from donor, facility, ministry Private payments by staff or patients Community Other (Specify)

		Don’t know Not applicable
Who installed the primary energy system?		Health ministry workers Christian Health Association Donor Health facility staff Collaboration of donor & ministry Private contractor Community Other (Specify) Don’t know Not applicable
Maintenance (preventive)	${electricity} = ‘yes’	
Is preventive maintenance carried out for the energy systems? **Preventive maintenance means the service is carried out even when there is no problem with the system.**		Routinely Sometimes but not routinely Never Not applicable
Is there a maintenance schedule for the energy system?		Yes No Don't know
Are maintenance records kept?		Yes, paper records Yes, digital records No Don’t know
Is there a service or maintenance contract in place for the energy system?		Yes No Don't know
Is the system under warranty?		Yes No Don't know
Who carries out the preventive maintenance for the energy systems?	Check all that apply	Designated facility maintenance staff Technicians from district or regional offices Technicians from Christian Health Association Donor Private contractor Other (Specify) Don’t know Not applicable
When was the energy system last serviced or maintained?		Within the past 3 months 3–6 months ago More than 6 months ago Never Don't know
Monitoring (predictive)	${electricity} = ‘yes’	
Is any part of the energy system remotely monitored?		Yes No Don’t know
What is remotely monitored?	${remote} = ‘yes’ Check all that apply	Central supply (national/community grid) Local mini-grid On-site hybrid solar/generator

		On-site generator (with battery storage) On-site generator (no battery storage) On-site solar system (except lanterns) Dedicated hydropower system Dedicated wind power system Rechargeable battery system Solar lanterns Other (Specify)
Repair (corrective)	${electricity} = 'yes'	
Has any part of the system been damaged in the last 12 months?		Yes No Don't know
What was damaged?	${damage} = 'yes' Check all that apply	Central supply (national/community grid) Local mini-grid On-site hybrid solar/generator On-site generator (with battery storage) On-site generator (no battery storage) On-site solar system (except lanterns) Dedicated hydropower system Dedicated wind power system Rechargeable battery system Solar lanterns Other (Specify)
How severe was the damage?	For each selection above	Complete breakdown Partial damage, but operational Superficial damage
Was the damage repaired?	For each selection above	Yes No Don't know
Who carried out the repairs?	${repair} = 'yes'	Ministry/Health Office Christian Health Association Donor Health facility staff Private contractor Don't know Not applicable
How long did you have to wait for the repair?	${repair} = 'yes'	Less than 24 hours 1-3 days 3-7 days 1-2 weeks 3-5 weeks Still not functional Don't know
Is there a designated staff member trained in managing or troubleshooting the energy system?		Yes No Partially Don't know
Do you have access to spare parts or replacements for the energy system?		Yes No Partially Don't know

Suitability and Future Needs		
In your opinion, is the current electricity system sufficient for your current needs?		Yes No Sometimes Don't know
Do you expect your electricity needs to increase in the next 12 months?		Yes No Don't know
Are there plans to upgrade or expand the electricity system?		Yes No Don't know

Climate Change & Weather Impact		
Question	**Logic**	**Choices**
Weather and Climate Change		
Have you observed any changes in weather patterns over the past few years (e.g. stronger storms, more heat, longer dry seasons)?		Yes No Don’t know
Which of the following weather changes have been most noticeable?	${weather} = ‘yes’	Increased temperatures Heatwaves More frequent storms Flooding Longer dry periods Other (specify)
How familiar are you with the concept of climate change?		Very familiar Somewhat familiar Heard of it, but not sure what it means Not familiar at all
Based on your current knowledge, what or who is the cause of climate change?		Human activities (e.g. burning fossil fuels, deforestation) Natural processes (e.g. volcanic activity, changes in solar radiation) Both human and natural causes Climate change is not happening I don’t know Other (specify)
In your opinion, is climate change affecting your community or the way health services are delivered?		Yes, significantly Yes, to some extent No Don’t know
Facility Impact		
Have changes in weather patterns (e.g., extreme heat, storms, floods) affected the operations at your facility?		Yes, significantly Yes, somewhat No Don’t know

What parts of facility operations have been most affected by extreme weather events?		Electricity reliability Water availability Staff/patient access to the facility Damage to buildings or equipment Other (specify)
During or after weather events, has energy availability (e.g., outages, damage to systems) affected patient care?		Yes, significantly Yes, somewhat No Don't know
Patient Population Impact		
During or after weather events, has there been any change in patient attendance at the facility?		Increase Decrease No change Don't know
Have you observed any changes in the patient population (e.g. more/less patients, more migrants, more seasonal illnesses like malaria, etc..) over the last two years?		Yes No Don't know
In your view, which of these changes are related to weather or environmental conditions?	${population} = 'yes'	Migration Displaced persons Not seeking care due to outages Delayed care-seeking due to transportation issues Increase in seasonal illnesses Reduced ability to refer patients to other facilities Other (specify)

Amenities & Equipment		
Question	**Logic**	**Choices**
Basic Amenities		
Is there a water source available?		Yes No
What does the water source of the facility look like?	Check all that apply	Piped Public tap Standpipe Tubewell/Borehole Protected Dug Well Protected Spring Rain Water
Where is the water source located?	single-choice	Inside facility Outside, within grounds Outside, external Other
Is there a separate, private consultation room available?		Yes No

Can a normal conversation be held in the consultation room without being overheard?		Yes No
Is there a toilet/latrine available for clients?		Yes No
What does the toilet/latrine look like?	Check all that apply	Flush/pour flush to piped sewer system or septic tank or pit latrine pit latrine (ventilated improved pit (VIP) or other) with slab composting toilet
Are there adequate sanitation facilities available on premises for clients?		Yes No
What functioning communication equipment is available?	Check all that apply	Private cell phone (only if reimbursed by facilitiy) Facility cell phone Other phone SW radio
Is there a functioning facility computer available?		Yes No
Does the computer have access to working internet? (day of survey)		Yes No
Is there an emergency transportation vehicle available for clients?		Yes; facility-owned, routinely available, with fuel Yes; access to vehicle near facility, routinely available, with fuel No

Basic Equipment	
Items observed and functioning in the main service area (usually the general outpatient department) or in the immediate vicinity where it is reasonable to assume that they can be used for the services being provided in the main service area	
Adult Scale	Yes No
Child Scale (250g steps min)	Yes No
Thermometer	Yes No
Stethoscope	Yes No
Blood pressure apparatus	Yes No
Light source (Spotlight/flashlight for examinations)	Yes No

Standard precautions for infection prevention		
How are sharps disposed of?	Check all that apply	Incinerator (functioning & fuel available) Open burning in protected area Dump without burning in protected area Remove offsite with protected storage Other None

Are unprotected sharps observed on site?		Yes No
How is infectious waste disposed of?	Check all that apply	Incinerator (functioning & fuel available) Open burning in protected area Dump without burning in protected area Remove offsite with protected storage Other None
Is unprotected infectious waste observed on site?		Yes No
Is storage for sharps observed on site?		Yes No
What does the storage for sharps look like?	Check all that apply	Puncture-resistant Rigid Leak-resistant Container to hold used sharps safely during collection, disposal and distruction Material: plastic Material: metal Material: cardboard Closeable For sharps of all standard sizes Clearly marked with intl. Biohazard warning no less than 50mm in diameter, printed in black or red on each of the front an back faces of the box Other None
Where was the storage for sharps observed?	Check all that apply	General OPD HIV testing area Surgery area Other None
Is there appropriate storage for infectious waste? (Waste receptacle (pedal bin) with lid and plastic bin liner)		Yes No
Is there disinfectant available anywhere in the facility? (Chlorine-based or other country specific used for environmental disinfection)		Yes No
Are there Single use —standard disposable or auto-disable syringes available anywhere in the facility?		Yes No
Where is Soap and running water or alcohol based hand rub available in the facilitiy?	Check all that apply	General OPD HIV testing area Basic obstetric area Newborn care area Surgery area Other None
Where are Latex gloves available in the facilitiy? (If equivalent non latex gloves are available this is acceptable.)	Check all that apply	General OPD HIV testing area Basic obstetric area Newborn care area Surgery area Other

	None
Are there guidelines for standard precautions available anywhere in the facility?	Yes No

Diagnostic capacity	
Able to conduct the test onsite (in the facility) and functioning equipment and reagents needed to conduct the test are observed onsite on the day of the survey. These may be in a laboratory or in the service area where the test is conducted.	
Haemoglobin (This may include colorimeter OR haemoglobinometer OR hemocue.)	Yes No
Blood glucose (Glucometer and glucometer test strips)	Yes No
Malaria diagnostic capacity (RDT kit or smear with microscope, slides, and Wright Giemsa stain)	Yes No
Urine dipstick- protein (Dipsticks for urine protein (with valid expiration date))	Yes No
Urine dipstick- glucose (Dipsticks for urine glucose (with valid expiration date))	Yes No
HIV diagnostic capacity (RDT kit or ELISA test with ELISA washer, ELISA reader, incubator, specific assay kit)	Yes No
Syphilis rapid test (RDT kit)	Yes No
Urine test for pregnancy (RDT kit)	Yes No

Essential Medicines	
Observed in pharmacy or in area where they are routinely stored, at least one with valid expiration date.	
Amlodipine tablet or alternative calcium channel blocker	Yes No
Amoxicillin syrup/suspension or dispersible tablet	Yes No
Amoxicillin tablet (Respiratory antibiotic)	Yes No
Ampicillin powder for injection	Yes No
Aspirin cap/tab	Yes No
Beclometasone inhaler	Yes No
Beta blocker (e.g.bisoprolol, metoprolol, carvedilol, atenolol)	Yes No
Carbamazepine tablet	Yes No
Ceftriaxone injection (2nd line injectable antibiotic)	Yes No
Diazepam injection	Yes No

Enalapril tablet or alternative ACE inhibitor e.g. lisinopril, ramipril, perindopril	Yes No
Fluoxetine tablet	Yes No
Gentamicin injection	Yes No
Glibenclamide tablet (Oral treatment type 2 diabetes)	Yes No
Haloperidol tablet	Yes No
Insulin regular injection	Yes No
Magnesium sulphate injectable	Yes No
Metformin tablet	Yes No
Omeprazole tablet or alternative such as pantoprazole, rabeprazole (Gastro-esophogeal reflux)	Yes No
Oral rehydration solution	Yes No
Oxytocin injection	Yes No
Salbutamol inhaler (Chronic asthma attacks)	Yes No
Simvastatin tablet or other statin e.g. atorvastatin, pravastatin, fluvastatin (High cholesterol)	Yes No
Thiazide (e.g. hydrochlorothiazide)	Yes No
Zinc sulphate tablets, dispersible tablets or syrup	Yes No

Appendix 3b: Questionnaire Chichewa

Facility Identifiers

Question	Logic	Choices
Dzina la chipatala ndi chiyani?		[Free text]
Kodi chipatalachi chili kuti kapena perekani kufotokozera malo ake?		[Free text]
Kutalika kuchokera pamwamba pa nyanja (m)		__ __ __ __
Latitude (M'lifupi mwa dziko)		N/S __ DEGREES __ __ DECIMAL __ __ __ __
Longitude (M'litali mwa dziko)		N/S __ DEGREES __ __ DECIMAL __ __ __ __
Malo ogwirira ntchitoyi ali m'chigawo/gawo chiti?		Kumpoto Chigawo Chapakati Kumwera
Malowa ali m'boma liti?		[List of all districts]
Ndi malo amtundu wanji awa?		chipatala chachikulu chotumizira odwala chipatala chigawo Chipatala chipatala chaching'ono. chipatala cha amayi/mwana China (nenani)
Ndani amayang'anira malo awa?		Boma Bungwe Losaphindula Bungwe lophindula Bungwe lachipembedzo (chikristu/chisilamu) China (nenani)
Kodi malo awa ali kudera lakumzinda kapena kumidzi?		Kumzinda Kumudzi
Kodi malo awa amangopatsa chithandizo odwala akunja okha?		Eya Ayi
"Pafupifupi, odwala angati amabwera ku malo awa patsiku?"		0 2025-01-10 2025-11-20 21 mpaka 50 51 mpaka 100 Kuposa 100
"Pafupifupi, odwala angati amabwera ku malo awa pasabata?"		0 1950-01-01 51 mpaka 100 101 mpaka 200 201 mpaka 500 Kuposa 500
Ndi masiku ati a sabata omwe malo awa amakhala otseguka kwa odwala?	Sankhani zonse zomwe zikugwirizana	Lolemba Lachiwiri Lachitatu Lachinayi Lachisanu

		Loweruka Lamulungu
"Pafupifupi, ndi masiku ati a sabata omwe pamakhala anthu ambiri?"	Sankhani zonse zomwe zikugwirizana	Lolemba Lachiwiri Lachitatu Lachinayi Lachisanu Loweruka Lamulungu
How long do outpatients stay in the clinic?		Kuchepela 1 Ola Maola 1 mpaka 2 Maola 2 mpaka 4 Maola 4 mpaka 6 Kuposa maola 6
Odwala akunja amakhala nthawi yayitali bwanji ku chipatala?		Kuchepela 5 mphindu 5 mpaka 15 mphindu 15 mpaka 30 mphindu 30 mpaka 60 mphindu Kuposa mphindu 60
Kodi odwala amapita bwanji ku chipatala?	Sankhani zonse zomwe zikugwirizana	Kuyenda Njinga Galimoto (yawo/taxi) Minibus (yagulu) Galimoto yaku chipatala China (nenani)
Kodi kuchuluka kwa odwala kumasintha bwanji chaka chonse?Kodi pamakhala nyengo yotanganidwa/ndi nyengo iti yomwe imakhala yotanganidwa kwambiri?	Sankhani zonse zomwe zikugwirizana	Nyengo yotentha komanso yamvula (Novembala mpaka Epulo) Nyengo yozizira komanso youma (Meyi mpaka Ogasiti) Nyengo yotentha komanso youma (Seputembala mpaka Okutobala) Nyengo yotentha komanso youma (Seputembala mpaka Okutobala)
"Energy System; SARA Energy Module (WHO, 2015)"		
Question	Logic	Choices
Questions on capacity		
"Kodi malo anu ali ndi magetsi ochokera kulikonse (mwachitsanzo, grid, jenereta, solar, kapena kwina)?"	Dumphani	Eya Ayi
Kodi magetsi amagwiritsidwa ntchito chiyani m'malo awa?	${electricity} = 'no'	"Zida zamagetsi zoyimirira zokha zokha (mwachitsanzo, chipinda choziziritsa cha EPUI, chida choyamwira, firiji, ndi zina zotero)."
	Sankhani zonse zomwe zikugwirizana	Kuwala kwamagetsi (kupatula nyale zamanzu) ndi zolumikizirana "Kuwala kwamagetsi, zolumikizirana, ndi zida zamankhwala chimodzi kapena ziwiri." Zofunikira zonse zamagetsi za malo awa Zofunikira zonse zamagetsi za malo awa ndi nyumba za ogwira ntchito

"Ngati malo awa alibe magetsi, ndi chiyani chomwe amapezera kuwala kwakukulu?"	Sankhani zonse zomwe zikugwirizana	Palibe gwero la kuwala Nyali za palafini Makandulo Nyali zogwiritsa ntchito mabatire Nyali za sola (dzuwa) Nyali zakumutu China (Nenani)
"Ngati malo awa ali ndi magetsi, ndi gwero liti lomwe amadalira kwambiri? Chonde dziwani: Izi zikutanthauza gwero la magetsi lomwe mumadalira nthawi zambiri."	${electricity} = 'yes'	Magetsi ochokera ku ESCOM adziko kappa dela lonse "Mini-grid yapafupi (kutanthauza kuti malo awa amagawana magetsi, nthawi zambiri ochokera ku gwero losinthika kapena jenereta, ndi nyumba zina zapafupi). Tchulani njira yoperekera magetsi ngati ikudziwika." Solar/jenereta yophatikizana pamalo pomwe Jenereta yapamalo pomwe (yokhala ndi chosungira magetsi mabatire) Jenereta yapamalo pomwe (yopanda chosungira magetsi mabatire) Dongosolo la solar lapamalo pomwe (kupatula nyali zanyumba) Dongosolo la magetsi ochokela ku madzi opatulika Dongosolo la magetsi ochokela ku mphepo opatulika China (Nenani)
Kodi malo awa ali ndi njira yosungira mphamvu zamagetsi monga mabatire oyendetsedwanso?		Eya Ayi Sindikudziwa
"Kupatula gwero lalikulu, kodi malo awa ali ndi gwero lina la magetsi loyimilira? Ngati lili nalo, ndi gwero liti lachiwiri la magetsi?"	${electricity} = 'yes'	Palibe gwero loyimilira
	Sankhani zonse zomwe zikugwirizana	Magetsi ochokera ku ESCOM adziko kappa dela lonse "Mini-grid yapafupi (kutanthauza kuti malo awa amagawana magetsi, nthawi zambiri ochokera ku gwero losinthika kapena jenereta, ndi nyumba zina zapafupi). Tchulani njira yoperekera magetsi ngati ikudziwika." Jenereta yapamalo pomwe (yokhala ndi chosungira magetsi mabatire) Jenereta yapamalo pomwe (yopanda chosungira magetsi mabatire) Dongosolo la solar lapamalo pomwe (kupatula nyali zanyumba) Dongosolo la magetsi ochokela ku madzi opatulika Dongosolo la magetsi ochokela ku mphepo opatulika Nyali za sola (dzuwa) China (Nenani)

"Ngati muli ndi magetsi akuluakulu ndi othandizira, kodi muli ndi njira yoyendetsera magetsi yosintha yokha?"	${secondary} = 'yes'	"Inde, kusintha kokha, ndiponso kuchepetsa kugwiritsa ntchito mafuta moyenera (mwachitsanzo, pogwiritsa ntchito mphepo/solar)." "Inde, kusintha kokha; koma palibe kuchepetsa kugwiritsa ntchito moyenera" "Ayi, kuyatsa pamanja kumafunika"
"Ngati mulibe magetsi oyimilira, ndi chiyani chomwe mumagwiritsa ntchito ngati gwero lachiwiri la kuwala?"	${secondary} = 'no'	Palibe gwero loyimilira la kuwala Nyali za palafini Makandulo Nyali zoyendetsedwa ndi batri Nyali zakumutu Nyali za sola (dzuwa) China (Nenani)
Generator Functionality		
Kodi jenereta ili ndi batire yodzaza lero?	selected(${generator})	Eya Ayi Sindikudziwa
Kodi jenereta ili ndi mafuta lero?	selected(${generator})	Eya Petroli Dizilo Ayi Sindikudziwa
Solar System Functionality		
Kodi dongosolo la solar likugwira ntchito?	selected(${solar})	"Inde, likugwira ntchito monga momwe likuyembekezeredwa" Pang'ono Battery ikufunika kukhozedwa kapena kusinthidwa Tchulani magawo onse "Ayi, sikugwira ntchito" Sindikudziwa
"Pafupifupi, dongosolo la solar limapereka magetsi maola angati patsiku?"	selected(${solar})	Maola 4 kapena kuchepera Maola 5 mpaka 8 9 mpaka 16 maola Maola 17 mpaka 23 (njira yosungira mphamvu zamagetsi ikufunika) Maola 24 (njira yosungira mphamvu zamagetsi ikufunika)
Duration		
"Pafupifupi, malo awa amakhala otseguka maola angati patsiku?"		Maola 4 kapena kuchepera Maola 5 mpaka 8 9 mpaka 16 maola Maola 17 mpaka 23 Maola 24
"Pafupifupi, m'maola 24 patsiku, magetsi amakhala alipo maola angati ngati mungafunike kuwagwiritsa ntchito?"		Maola 4 kapena kuchepera Maola 5 mpaka 8 Maola 9 mpaka 16 Maola 17 mpaka 23 (njira yosungira mphamvu

		zamagetsi ikufunika) Maola 24
Evening supply		
"Pafupifupi, malo awa amakhala otseguka maola angati madzulo (pambuyo pa 6 koloko masana) ndi usiku kuti apereke chithandizo?"		Samatseguka konse pambuyo pa 6 koloko masana Mpakna 8 koloko madzulo Mpakana 10 koloko madzulo Mpakana pakati pa usiku Usiku onse
"Pafupifupi, magetsi amakhala alipo maola angati madzulo (pambuyo pa 6 koloko masana) ndi usiku pa malo a zaumoyo?"		Ochepera ola limodzi 1 Ola mpaka Maola 2 Maola 3 mpaka 4 Maola 5 mpaka 8 Maola 8 mpaka 14
Reliability		
"M'masiku 7 apitawa, kodi magetsi analipo nthawi zonse pamene amafunikira kuchokera ku gwero lalikulu kapena loyimilira lililonse pamene malo awa anali otseguka kuti apereke chithandizo chokhazikika kapena cha ngozi?"		Nthawi zonse alipo (palibe zododometsa) Nthawi zambiri alipo (kudodometsa kwakanthawi kochepera maola awiri patsiku) Nthawi zina alipo (kudodometsa pafupipafupi kapena kwautali wopitilira maola awiri patsiku)
"Ngati malo awa ali olumikizidwa ku magetsi a ESCOM, pafupifupi, kangati patsiku amakumana ndi kuthima kwa zamagetsi kuchokera ku ESCOM?"		Sama thima Amathima kamodzi mpaka katatu patsiku Amathima kuposa katatu patsiku
"Ngati magetsi amachokera makamaka ku jenereta yapamalo pomwe, pafupifupi, kangati patsiku malo awa amakumana ndi zododometsa zamagetsi?"		Palibe zododometsa Kudodometsa 1 mpaka 3 patsiku Kudodometsa kopitilira 3 patsiku
"Pafupifupi, patsiku, kodi nthawi zonse zododometsa za magetsi zikaphatikizidwa zingakhale nthawi yochuluka bwanji?"		Palibe zododometsa Ochepera mphindi 30 Mphindi 31 mpaka 60 Mphindi 61 mpaka maola 4 Maola 5 mpaka 8 Kupitilira Maola 8
Kodi zododometsa zamagetsi zimachitika kwambiri nthawi zanji?		Kukacha mpaka Masana (kusanade) Madzulo (Kukada mpaka pakati pa usiku) Usiku (pakati pa haiku mpaka koloko m'masa)
Quality		
Kodi malo awa nthawi zambiri amakumana ndi vuto la magetsi ochepa (low voltage) kapena kusakhazikika kwa magetsi (voltage fluctuations) kuchokera ku gwero lalikulu?		Eya Ayi Sindikudziwa
Kodi kusakhazikika kwa magetsi (voltage fluctuations) kunawononga zida zilizonse m'chaka chatha?		Ayi Zida zina zinawonongeka Zida zambiri zinawonongeka
Financial Sustainability		
"Ngati mumagwiritsa ntchito magetsi ochokera ku grid, kodi mudakhala ndi ndalama zokwanira zolipirira ma bill anu amagetsi m'miyezi iwiri yapitayi?"	selected(${grid})	Eya Ayi Sindikudziwa Sizikugwirizana

"Ngati mumagwiritsa ntchito magetsi ochokera ku grid, ndani amamalipira bill yamagetsi ya chipatala chaching'ono cha zaumoyo?"	Sankhani zonse zomwe zikugwirizana	Unduna wa Zaumoyo/DHO Wopereka chithandizo Malo a zaumoyo "Ndalama zophatikizidwa kuchokera kwa wopereka chithandizo, malo a zaumoyo, ndi unduna" Odwala Sindikudziwa
"M'miyezi iwiri yapitayi, kodi mudakhala ndi ndalama zokwanira (ndalama zomwe zilipo) zogulira mafuta a jenereta yoyimilira kuti igwire ntchito nthawi zonse pamene inkafunika?"	selected(${generator})	Eya Ayi Sindikudziwa
Ndani amalipira mafuta a jenereta?	selected(${generator})	Unduna wa Zaumoyo/DHO
	Sankhani zonse zomwe zikugwirizana	Wopereka chithandizo Malo a zaumoyo "Ndalama zophatikizidwa kuchokera kwa wopereka chithandizo, malo a zaumoyo, ndi unduna" Magetsi onse amalembedwa mita ndikulipiritsidwa odwala Magetsi ena amalipiritsidwa odwala Sindikudziwa
"M'chaka chatha, kodi mudakhala ndi ndalama zokwanira (ndalama zomwe zilipo) zosungira mapanelo a solar akugwira ntchito"	selected(${solar})	Eya Ayi Sindikudziwa
"Ngati mumagwiritsa ntchito njira yamagetsi yochokera ku solar, ndani amalipira mababu a magetsi ndi zida zina zosinthira za solar?"	selected(${solar})	Unduna wa Zaumoyo/DHO
	Sankhani zonse zomwe zikugwirizana	Wopereka chithandizo Malo a zaumoyo Odwala "Ndalama zophatikizidwa kuchokera kwa wopereka chithandizo, malo a zaumoyo, ndi unduna" Malipiro apadera ochokera kwa antchito kapena odwala Kuphatikiza kugula kwaumwini ndi magwero ena Palibe Sindikudziwa
"M'miyezi iwiri yapitayi, kodi mudakhala ndi ndalama zokwanira (ndalama zomwe zilipo) zosamalira zida zoperekera magetsi?"	${electricity} = 'yes'	Eya Ayi Sindikudziwa
Environmental Sustainability		
"Kodi gwero lanu lalikulu la kuwala (kuphatikizapo nyale za parafini kapena makandulo) kapena magetsi limayambitsa kuwonongeka kwa mpweya m'dera lanu mkati kapena mozungulira chipatala chaching'ono cha zaumoyo, kudzera mu kutulutsa utsi kapena mpweya		Eya Ayi Sindikudziwa

umene ungathe kuwononga thanzi la odwala kapena antchito a zaumoyo?"		
"Kodi gwero lanu lalikulu la kuwala kapena magetsi limayambitsa kuwonongeka kwa madzi kapena nthaka m'dera lanu mkati kapena mozungulira chipatala chaching'ono cha zaumoyo, kudzera mu kutulutsa mafuta, utsi, kapena zinyalala zina zomwe zingawononge thanzi la odwala kapena antchito a zaumoyo?"		Eya Ayi Sindikudziwa
Kodi gwero lanu lalikulu la magetsi limayambitsa phokoso lalikulu m'dera lanu mkati kapena mozungulira chipatala chaching'ono cha zaumoyo lomwe lingawononge thanzi la odwala kapena antchito a zaumoyo?		Eya Ayi Sindikudziwa
"Origins, Operations and Maintenance"		
Question	Logic	Choices
Origins	${electricity} = 'yes'	
Ndani adalipira ndalama zoyambira kukhazikitsa njira yoyamba yamagetsi pa malo awa?		Unduna wa Zaumoyo/DHO Bungwe la Zaumoyo la Chikhristu Wopereka chithandizo Malo a zaumoyo "Ndalama zophatikizidwa kuchokera kwa wopereka chithandizo, malo a zaumoyo, ndi unduna" Malipiro apadera ochokera kwa antchito kapena odwala Mudzi China (nenani) Sindikudziwa Sizikugwirizana
Ndani adaika njira yoyamba yamagetsi?		Antchito a Unduna wa Zaumoyo Bungwe la Zaumoyo la Chikhristu Wopereka chithandizo Antchito ku Malo a zaumoyo Mgwirizano pakati pa wopereka chithandizo ndi unduna Kontrakitala Mudzi China (nenani) Sindikudziwa Sizikugwirizana
Maintenance (preventive)	${electricity} = 'yes'	
IKodi ntchito yosamalira makina a magetsi imachitika nthawi zonse pofuna kupewa kuwonongeka?		Nthawi zonse Nthawi zina koma osati nthawi zonse Sanapangepo Sizikugwirizana
Kodi pali ndondomeko yosamalira makina amagetsi?		Eya Ayi Sindikudziwa

Kodi zolemba zosamalira (maintenance records) zimasungidwa?		"Inde, zolemba zapapepala" "Inde, zolemba za digito" Ayi Sindikudziwa
Kodi pali mgwirizano wa ntchito kapena wosamalira makina amagetsi?		Eya Ayi Sindikudziwa
Kodi makinawo ali pansi pa chitsimikizo (warranty)?		Eya Ayi Sindikudziwa
Ndani amachita ntchito yosamalira makina amagetsi pofuna kupewa kuwonongeka?	Sankhani zonse zomwe zikugwirizana	Antchito osamalira malo apadera Akatswiri ochokera ku maofesi a chigawo kapena zigawo Akatswiri ochokera ku Bungwe la Zaumoyo la Chikhrist Wopereka chithandizo Kontrakitala China (nenani) Sindikudziwa Sizikugwirizana
Kodi makina amagetsi adasamaliridwa kapena adakonzedwa liti komaliza?		M'miyezi itatu yapitayi Miyezi 3 mpaka 6 yapitayo Miyezi yoposa 6 yapitayo Sanapangepo Sindikudziwa
Monitoring (predictive)	${electricity} = 'yes'	
Kodi gawo lililonse la makina amagetsi likuyang'aniridwa kutali?		Eya Ayi Sindikudziwa
Kodi chomwe chikuyang'aniridwa kutali ndi chiyani?	${remote} = 'yes'	Magetsi ochokera ku boma Magetsi apaderadera
	Sankhani zonse zomwe zikugwirizana	Makina ophatikizira solar ndi jenereta pa malo omwe magetsi akufunikira Jenereta pa malo (yokhala ndi chosungira magetsi/batri Jenereta pa malo (wopanda chosungira magetsi/batri) Makina a solar pa malo (kupatula nyale) Makina apadera opangira magetsi kuchokera kumadzi Makina apadera opangira magetsi kuchokera kumphepo Makina a batri oyambiritsidwanso Nyali za sola (dzuwa) China (nenani)
Repair (corrective)	${electricity} = 'yes'	
Has any part of the system been damaged in the last 12 months?		Eya Ayi Sindikudziwa

Chiyani chinawonongeka?	${damage} = ‘yes’	Magetsi ochokera ku boma Magetsi apaderadera
	Sankhani zonse zomwe zikugwirizana	Makina ophatikizira solar ndi jenereta pa malo omwe magetsi akufunikira Jenereta pa malo (yokhala ndi chosungira magetsi/batri) Jenereta pa malo (wopanda chosungira magetsi/batri) Makina a solar pa malo (kupatula nyale) Makina apadera opangira magetsi kuchokera kumadzi Makina apadera opangira magetsi kuchokera kumphepo Makina a batri oyambiritsidwanso Nyali za sola (dzuwa) China (nenani)
Kodi kuwonongekako kunali koopsa motani?	For each selection above	Onse anawonongeka "Zinawonongeka pang'ono, koma zikugwirabe ntchito" Kuwonongeka kwapang'ono
Kodi kuwonongekako kunakonzedwa?	For each selection above	Eya Ayi Sindikudziwa
Ndani adakonza zowonongekazo?	${repair} = ‘yes’	Unduna wa Zaumoyo/DHO Bungwe la Zaumoyo la Chikhristu Wopereka chithandizo Antchito a Unduna wa Zaumoyo Kontrakitala Sindikudziwa Sizikugwirizana
Munadikirira nthawi yayitali bwanji kuti akonze zowonongekazo?	${repair} = ‘yes’	Zosakwana maola 24 Masiku 1 mpaka 3 Masiku 3 mpaka 7 Masabata 1 mpaka 2 Masabata 3 mpaka 5 Silikugwirabe ntchito Sindikudziwa
Kodi pali antchito apadera ophunzitsidwa kuyendetsa kapena kukonza mavuto a makina amagetsi?		Eya Ayi Pang'ono Sindikudziwa
Do you have access to spare parts or replacements for the energy system?		Eya Ayi Pang'ono Sindikudziwa
Suitability and Future Needs		
"Malingana ndi maganizo anu, kodi dongosolo lamagetsi lomwe lilipo tsopano likukwanira zosowa zanu zamakono?"		Eya Ayi Nthawi zina Sindikudziwa

Kodi mukuyembekezera kuti zosowa zanu za magetsi ziwonjezeka m'miyezi 12 ikubwerayi?		Eya Ayi Sindikudziwa
Kodi pali mapulani oti akweze kapena kukulitsa dongosolo la magetsi?		Eya Ayi Sindikudziwa
Climate Change & Weather Impact		
Question	Logic	Choices
Weather and Climate Change		
"Kodi mwawona kusintha kulikonse kwa nyengo m'zaka zingapo zapitazi (monga mvula yamphamvu, kutentha kwambiri, kapena nyengo youma yayitali)?"		Eya Ayi Sindikudziwa
Ndi kusintha kwanthawi yotani komwe kwakhala koonekeratu kwambiri?	${weather} = 'yes'	Kutentha kwawonjezeka Mafunde otentha Mvula zamphamvu zambiri Kusefukira kwa madzi Nyengo youma yayitali China (nenani)
Kodi mukudziwa bwino bwanji za lingaliro la kusintha kwa nyengo?		Ndikudziwa bwino kwambir Ndikudziwa pang'ono "Ndamvapo, koma sindikudziwa tanthauzo lake" Sindikudziwa konse
"Kutengera chidziwitso chanu chamakono, kodi chomwe chimayambitsa kusintha kwa nyengo ndi chiyani kapena ndani?"		"Zoyendera za anthu (mwachitsanzo, kuyatsa mafuta, kudula mitengo)." "Zochitika zachilengedwe (mwachitsanzo, kuphulika kwa mapiri, kusintha kwa mphamvu ya dzuwa)." Zomwe zimayambitsidwa ndi anthu komanso zachilengedwe Kusintha kwa nyengo sikukuchitika Sindikudziwa China (nenani)
"Malingana ndi maganizo anu, kodi kusintha kwa nyengo kukukhudza madera anu kapena momwe ntchito za zaumoyo zikuperekedwera?"		"Eya, Kwambiri " "Eya, Pang'ono " Ayi Sindikudziwa
Facility Impact		
"Kodi kusintha kwa nyengo (mwachitsanzo, kutentha kwambiri, mvula yamphamvu, kusefukira kwa madzi) kwakhudza ntchito zoyendetsera chipatala chanu?"		"Eya, Kwambiri " "Eya, Pang'ono " Ayi Sindikudziwa
Ndi magawo ati a ntchito za chipatala omwe akhudzidwa kwambiri ndi nyengo yoipa kwambiri?		Kupezeka kwa magetsi Kupezeka kwa madzi Kufikika kwa antchito/odwala ku chipatala Kuwonongeka kwa nyumba kapena zipangizo China (nenani)
"Pa nthawi kapena pambuyo pa masoka achilengedwe, kodi kupezeka kwa magetsi (mwachitsanzo, kudulika		"Eya, Kwambiri " "Eya, Pang'ono "

kwa magetsi, kuwonongeka kwa makina) kwakhudza chithandizo cha odwala?"		Ayi Sindikudziwa
Patient Population Impact		
"Pa nthawi kapena pambuyo pa masoka achilengedwe, kodi pakhala kusintha kulikonse kwa chiwerengero cha odwala obwera ku chipatala?"		Kuchuluka Kuchepa Palibe kusintha Sindikudziwa
"Kodi mwawona kusintha kulikonse pa chiwerengero cha odwala (mwachitsanzo, odwala ochuluka/ochepa, ochoka kwawo ambiri, matenda a nyengo ochuluka monga malungo, ndi zina zotero) pazaka ziwiri zapitazi?"		Eya Ayi Sindikudziwa
"Malingana ndi maganizo anu, ndi kusintha kotani mwa izi komwe kukugwirizana ndi nyengo kapena zinthu za chilengedwe?"	${population} = 'yes'	Kusuntha kwa anthu Anthu osamutsidwa Osafuna chithandizo chifukwa cha kusakhalapo kwa magetsi Kuchedwa kufuna chithandizo chifukwa cha mavuto oyendera Kuchuluka kwa matenda a nyengo Kuchepa kwa mwayi wotumiza odwala kumipatala ina China (nenani)
Amenities & Equipment		
Question	Logic	Choices
Basic Amenities		
Kodi pali gwero la madzi lomwe likupezeka?		Eya Ayi
Kodi gwero la madzi la chipatala likuwoneka bwanji?	Sankhani zonse zomwe zikugwirizana	Madzi a mapaipi Pampopi pa anthu onse Chitoliro choyima Chitsime chobowola Chitsime chokumbidwa chotetezedwa Kasupe Wotetezedwa Madzi a Mvula
Kodi gwero la madzi lili kuti?	single-choice	Mkati mwa chipatala "Panja, mkati mwa mpanda wa chipatala" "Panja, pa chipatala" China (nenani)
Kodi pali chipinda chapadera chobisika chokambilana nacho?		Eya Ayi
Kodi zotheka kukambirana mwachinsinsi mu chipinda chokambilana popanda kumvedwa ndi wina?		Eya Ayi
Kodi pali chimbudzi chomwe chilipo choti makasitomala agwiritse ntchito?		Eya Ayi
Kodi chimbudzi chikuperekeka bwanji?	Sankhani zonse zomwe zikugwirizana	Chimbudzi chotsuka/chothira madzi chopita ku mapaipi akuluakulu otulutsa zonyansa kapena septic tank kapena dzenje la chimbudzi. Chimbudzi cha dzenje (chokhala ndi mpweya wabwino (VIP) kapena china) chokhala ndi

		nthaka/nthanda Chimbudzi chopangira manyowa
Kodi pali zimbudzi zokwanira zomwe zilipo paderalo kuti makasitomala agwiritse ntchito?		Eya Ayi
Kodi zida zoyankhulirana zomwe zikugwira ntchito zilipo ziti?	Sankhani zonse zomwe zikugwirizana	Foni yanu yam'manja (ngati chipatala chikubwezerani ndalama zake zokha) Foni ya chipatala Foni ina Wayilesi ya mawu aatali
Pali kompyuta ya chipatala yomwe ikugwira ntchito?		Eya Ayi
Kodi kompyutayo ili ndi intaneti yogwira ntchito? (tsiku la kafukufuku)		Eya Ayi
Kodi pali galimoto yadzidzidzi yonyamula odwala?		"Inde; chipatala, chimapezeka nthawi zonse, chili ndi mafuta" "Inde; galimoto ili pafupi ndi bungwe, imapezeka nthawi zonse, ndipo ili ndi mafuta" Ayi
Basic Equipment		
Zinthu zomwe zawonedwa ndipo zikugwira ntchito m'dera lalikulu la ntchito (nthawi zambiri chipinda chachipatala cha odwala akunja) kapena m'dera loyandikana nalo momwe tingaganizire kuti zingagwiritsidwe ntchito pothandiza ntchito zomwe zikuchitika m'dera lalikululi		
Scelo ya kulemera kwa akuluakulu		Eya Ayi
Scelo ya kulemera kwa ana		Eya Ayi
Thermometa		Eya Ayi
Sitetosikopu		Eya Ayi
Chida cha kuthamanga kwa magazi		Eya Ayi
chowunikira pa kafukufuku wa thanzi		Eya Ayi
Standard precautions for infection prevention		
Zinthu zolimba zokhudzana ndi thanzi zimachotsedwa bwanji?	Sankhani zonse zomwe zikugwirizana	"Malo owotsera zinyalala (akugwira ntchito, mafuta alipo)" Kutentha moto m'malo otetezedwa Kutaya zinyalala popanda kutentha moto m'malo otetezedwa Kutengera kunja kwa malo ndi kusungira bwino China Palibe
Kodi zida zolimba zosakhudzidwa bwino zowonetsa pa malo?		Eya Ayi
Zinyalala zodziwikiratu za matenda zimachotsedwa bwanji?	Sankhani zonse zomwe zikugwirizana	"Malo owotsera zinyalala (akugwira ntchito, mafuta alipo)"

		Kutentha moto m’malo otetezedwa Kutaya zinyalala popanda kutentha moto m’malo otetezedwa Kutengera kunja kwa malo ndi kusungira bwino China Palibe
Kodi chinyalala chotengera matenda chosadziwika bwino ndi chowonetsedwa pa malo?		Eya Ayi
Kodi malo osungira zida zolimba amaonekera pa malo?		Eya Ayi
Malo osungira zida zolimba amawoneka bwanji?	Sankhani zonse zomwe zikugwirizana	Chosagwira zovulaza Wolimba Wosatulukira madzi kapena zinthu zina "Chidebe chotetezera zida zolimba zogwiritsidwa ntchito pa kusonkhanitsa, kutayidwa ndi kuwonongeka" pulasitiki chitsulo Katoni Chotsekedwa Pazida zolimba za mitundu yonse yachikhalidwe "Zolembedwa bwino ndi chizindikiro cha chenjezo la Biohazard lachigawo cha dziko lonse, cholemera osachepera 50mm, cholimbikira chotsalira chakuda kapena chofiira patsogolo ndi kumbuyo kwa bokosi" China Palibe
Malo osungira zida zolimba adawonetsedwa kuti ali kuti?	Sankhani zonse zomwe zikugwirizana	Dipatimenti ya Odwala Wamba (OPD) Malo oyezela HIV Malo a opaleshoni China Palibe
Kodi pali malo oyenera osungira chinyalala chotengera matenda? (M'baketi yokhala ndi mpando wophatikizika ndi chivundikiro komanso layini ya pulasitiki)		Eya Ayi
Kodi pali mankhwala ochotsa tizilombo pa malo aliwonse m'bungweli? (Mankhwala olimbikitsidwa ndi klorini kapena ena omwe amagwiritsidwa ntchito m'malo ogwirira ntchito)		Eya Ayi
Kodi pali masilinji opangidwa kuti agwiritsidwe ntchito kamodzi kokha — a mtundu wotsalira kapena omwe amalephera kugwiritsidwanso ntchito ali kulikonse m’bungwe?		Eya Ayi
Kodi sopo ndi madzi oyenda kapena mankhwala osamba manja omwe ali ndi alukolo ali kuti kuchipatala?	Sankhani zonse zomwe zikugwirizana	Dipatimenti ya Odwala Wamba (OPD) Malo oyezela HIV Malo oyambira za umoyo wa amayi ndi ana Malo oyang'anira ana oadwa kumene Malo a opaleshoni China Palibe

"Kodi magolovesi a Latex ali kuti kuchipatala? (Ngati pali magolovesi ena osati a Latex omwe ali ofanana, izi zimavomerezeka.)"	Sankhani zonse zomwe zikugwirizana	Dipatimenti ya Odwala Wamba (OPD) Malo oyezela HIV Malo oyambira za umoyo wa amayi ndi ana Malo oyang'anira ana oadwa kumene Malo a opaleshoni China Palibe
Kodi pali malamulo kapena njira zowongolera zachilendo zomwe zimatsatira mawu a chitetezo zilipo kulikonse kuchipatala?		Eya Ayi
Diagnostic capacity		
Kutha kuchita mayeso pa malo (kuchipatala) ndipo zida ndi mankhwala ogwiritsira ntchito mayesowo zikupezeka ndipo zikugwira ntchito pa tsiku la kafukufuku. Izi zingakhale m'labotale kapena m'dera la ntchito komwe mayesowo amachitidwa.		
"Haemoglobin (Izi zingaphatikize colorimeter, haemoglobinometer, kapena hemocue."		Eya Ayi
Shuga m'magazi (Glucometer ndi mitundu ya mayeso a glucometer)		Eya Ayi
"Mphamvu yodziwira matenda a malungo (kit ya RDT kapena kusinkhasinkha ndi maikrosikopu, masilayidi, ndi mankhwala a Wright Giemsa)"		Eya Ayi
Dipstiki ya mikodzo – protein (Dipstiki za kuyesa protein mikodzo (zili ndi tsiku lololera lomwe likugwira ntchito))		Eya Ayi
Dipstiki ya mikodzo – shuga (Dipstiki za kuyesa shuga m'mikodzo (zili ndi tsiku lololera lomwe likugwira ntchito))		Eya Ayi
"Mphamvu yodziwira HIV (kit ya RDT kapena mayeso a ELISA ndi makina a kuyeretsa ELISA, wowerenga wa ELISA, incubator, ndi kit yapadera ya mayeso)"		Eya Ayi
Mayeso achangu a syphilis (kit ya RDT)		Eya Ayi
Mayeso a mikodzo a mimba (kit ya RDT)		Eya Ayi
Essential Medicines		
"Zawonedwa mufamu kapena m'dera lomwe zinthu zimakhala zikusungidwa nthawi zonse, zokwana chimodzi chokhala ndi tsiku lololera lomwe likugwira ntchito."		
Mapiritsi a Amlodipine kapena mankhwala ena oyang'anira calcium channel		Eya Ayi
Siraphu kapena suspenshoni ya Amoxicillin kapena mapiritsi omwe amatha kusungidwa m'madzi		Eya Ayi
Mapiritsi a Amoxicillin (mankhwala a matenda opuma)		Eya Ayi
Ufa wa Ampicillin wa mkingo		Eya Ayi

mapiritsi a Aspirin	Eya Ayi
Inhaler ya Beclometasone	Eya Ayi
"Mankhwala a Beta blocker (mwachitsanzo bisoprolol, metoprolol, carvedilol, atenolol)"	Eya Ayi
Piritsi la Carbamazepine	Eya Ayi
Mkingo wa Ceftriaxone (antibiotic ya mzere wachiwiri)	Eya Ayi
Mkingo wa Diazepam	Eya Ayi
"Mapiritsi a Enalapril kapena mankhwala ena a ACE inhibitor, mwachitsanzo lisinopril, ramipril, perindopril"	Eya Ayi
Piritsi la Fluoxetine	Eya Ayi
Mkingo wa Gentamicin	Eya Ayi
Piritsi la Glibenclamide (mankhwala a shuga wa mtundu wachiwiri)	Eya Ayi
Piritsi la Haloperidol	Eya Ayi
Mkingo wa Insulin wamba	Eya Ayi
Mkingo wa Magnesium sulphate	Eya Ayi
Piritsi la Metformin	Eya Ayi
"Piritsi la Omeprazole kapena mankhwala ena monga pantoprazole, rabeprazole (matenda obwera chifukwa cha asidi m'mimba kubwerera m'mphuno kapena m'mphuno mwa m’mimba)"	Eya Ayi
Madzi a mchere a m’kamwa obwezeretsa madzi m’thupi	Eya Ayi
Mkingo wa Oxytocin	Eya Ayi
Inhaler ya Salbutamol (matenda a asthma omwe amakhala nthawi yayitali)	Eya Ayi
"Piritsi la Simvastatin kapena mankhwala ena a statin monga atorvastatin, pravastatin, fluvastatin (cholesterol waukulu m’magazi)"	Eya Ayi
Thiazide (mwachitsanzo hydrochlorothiazide)	Eya Ayi
"Mapiritsi a zinc sulphate, mapiritsi omwe amasungunuka m'madzi kapena sirapu"	Eya Ayi

Appendix 4: Letter of Recommendation

OFFICE OF THE HEAD OF DEPARTMENT

AGRISCIENCES

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6/6/2025

Dear DHO,

I write to request that you provide access to your health facilities to Gian Teufen and Till Haeussner who are working with me on a health clinic electrification project.

Specifically, Mr. Teufen will be focused on developing an energy systems, operations and maintenance dataset of Malawi's health posts in which facility-level data on the energy supply status will be collected. This means documenting the energy source mix and energy storage by writing down nameplate characteristics and inspecting functionality and installation. Additionally, the origins and current operations and maintenance of energy systems and key medical devices will be documented by talking to the responsible persons. **The goal is then for this dataset to enable accountability among associated donors or implementing partners for implemented energy systems**.

Mr. Haeussner will be developing an open database of electrical medical devices for health posts in Malawi. Appropriately sizing a renewable energy system for healthcare facilities requires estimates on the energy load of the electrical devices present. The goal of this work is to start an open database of the electrical devices being used in Malawi's health posts. For this, data on the electrical device inventory will be collected, including brand, name, energy characteristics and functionality. Additionally, the power consumption profile of certain devices will be measured using a conventional energy logger. **The results of this research will guide more accurate load estimations in healthcare facility electrification efforts throughout Malawi and beyond.**

Thank you for your consideration,

Dr. Frank Mnthambala

Head of Agrisciences Department.